



Manipulating data

Consuming data



Reading data

To extract data, make an http GET request

Syntax:

http://[host]:[port]/semarchy/api/rest/query/[dataloc]/[entity]/[view_type]

Values for view_type:

SD	SE
MD	GE
GD	SDWE
GDWE	

REST reading errors

- .../[entity]/SE returns a record for each error
- .../[entity]/SDWE returns each record that has an error
- Same for GE/GDWE

SE:

```
{  
  "records" : [ {  
    "CustomerID" : 1001,  
    "Name" : "Husky Co",  
    "Phone" : null,  
    "Email" : null  
  }, {  
    "CustomerID" : 1001,  
    "Name" : "Husky Co",  
    "Phone" : null,  
    "Email" : null  
  } ]  
}
```

SDWE:

```
{  
  "records" : [ {  
    " CustomerID " : 1001,  
    "Name" : "Husky Co",  
    "Phone" : null,  
    "Email" : null  
  } ]  
}
```

Reading data - Parameters

Parameters available:

- Pagination
- Query expressions
- Filtering
- Sorting

Pagination

Pagination

\$offset (default 0)

\$limit (default 100)

E.g.:

Retrieve 5 countries at a time, third query

.../semarchy/api/rest/query/RDM/Country/GD?\$offset=10&\$limit=5

Query expressions

- Query expressions define which fields are returned
- \$baseexprs
 - USER_ATTRS - All attributes you defined (no system attributes, no FKs)
 - VIEW_ATTRS - All attributes you defined plus all system attributes (no FKs)
 - NONE
- \$expr
 - Add or exclude fields

E.g.:

.../Country/GD?\$baseexprs=NONE&\$expr=country:CommonName&\$offset=10&\$limit=5

Sorting and filtering

- Sorting and filtering
 - \$orderBy
 - \$f

- Single Record

http://[host]:[port]/semarchy/api/rest/query/[dataloc]/[entity]/[view_type]/[record_id]

`.../semarchy/api/rest/query/ReferenceData/Country
/GD?$f=CommonName%20like%20%27F%25%27%20AND%20Iso4217CurrencyCode=%27EUR%27`

Without URL encoding:
CommonName like 'F%' and
Iso4217CurrencyCode='EUR'

```
{  
  "records" : [ {  
    "country" : "Finland"  
  }, {  
    "country" : "France"  
  } ]  
}
```

Counting records

Get a count without returning the the data

`http://[host]:[port]/semarchy/api/rest/count/[dataloc]/[entity]/[view_type]`

`.../semarchy/api/rest/count/ReferenceData/Country
/GD?$f=CommonName%20like%20%27F%25%27%20AND%20Iso4217CurrencyCode=%27EUR%27`

Exactly the same query except using
'count' rather than 'query'

```
{  
  "count" : 2  
}
```


Download OpenAPI

From the REST API documentation you can download the *OpenAPI 3.0.0* description of each domain. The description file is usually consumed by integration tools to reverse-engineer the structure of the API

